



Anything but that!

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Bloom XB-1 airline

American startup Bloom plans to bring supersonic flight back to commercial passenger air travel with their new XB-1 <http://digit.in/BloomXB1>

Space Age

Cocconi speculated this particular frequency region (read here: <http://digit.in/SFICPaper>). Every single body in the universe consisting of mass generates radio waves, including us (human beings). Hydrogen is the most abundant element found in the universe, implying that most of the inhabitants must be familiar with it. Neutral Hydrogen emits radio waves at a frequency of 1,420 MHz. Assuming an equivalent level of intelligence, extra-terrestrials trying to establish contact with fellow inhabitants of the universe should have developed receivers capable of picking up signals in this range. Although a signal from a source might originate at the speculated frequency region, it could suffer attenuation while traversing due to several factors. To eliminate these factors, a signal with radio waves around the 1,420 MHz region (plus or minus 300 KHz) is to be considered as well. The paper was published almost a decade before we detected the Wow! Signal, which gave the paper more credibility. The reason was the two different reviewed data values of the signal's frequency – 1420.36 MHz and 1420.46 MHz – was close to the speculated one.

While trying to collect deep space signals, it is also necessary to eliminate known sources of signals which are not extra-terrestrial in nature. As already mentioned above that everything in the universe emits radio waves, sources such as planets, asteroids, satellites, aircrafts, space crafts and even ground-stationed transmitters have to be ruled out. In scientific terms, ruled out doesn't imply eliminated but something which has a low probability. Consider them as a checklist of sort including external factors such as gravitational lensing (a phenomenon that causes signals to bend around objects resulting in a brighter images) and interstellar scintillation (the twinkling effect). If these factors are ruled out, then the most probable cause of the deep space signal has to be from an foreign source.

The Wow! Signal lasted for 72 seconds generating an alphanumeric sequence "6EQUJ5" denoting the intensity variation of the radio signal. One of the instances in the Wow! Signal was almost 30 times stronger than normal deep space.

SETI

Most of the research and study to evaluate these mysterious signals are being led by SETI. A highly popular and old non-profit organisation, the SETI institute was incorporated in 1984. It mainly consists of three primary centres – Carl Sagan Centre, Centre for Education and the Centre for Public Outreach. The Carl Sagan Centre is dedicated for studying life in the universe, the Centre of Education deals with astronomy, astrobiology and space science while the Centre for Public Outreach takes care of the institute's general science radio show and podcast – Big Picture Science, and "SETI Talk" which is their weekly series.

Detection of signals are carried out using ground-based telescopes located at several observatories across USA. The main instruments used includes the ground-based Allen Telescope Array near San Francisco. They also employ ground-based optical telescopes like the Shane telescope at Lick Observatory, the W.M. Keck telescopes and IRTF in Hawaii, and the Very Large Telescopes in Chile. Research is also carried out using space telescope facilities such as the Hubble Space Telescope, the Spitzer Space Telescope and the Herschel Space Telescope.

Along with searching the universe for extra-terrestrials, SETI also consists of a project called Active SETI (Active Search for Extra-Terrestrial Intelligence) to send messages to intelligent life forms through radio signals. It's also called (METI) Messaging to Extra-Terrestrial Intelligence but the distinction lies in SETI doing only the searching of message whereas METI is more concerned with the creation of messages. We'll be talking more about Active SETI in the last section of this article.

Recent developments

So, what has suddenly piqued our interest into one of the most significant event in radio astronomy? For start-

ers, this year we'll be celebrating the 40th anniversary of the Wow! Signal while anticipating a similar event of such a scale to occur once again. Radio astronomers have been riddled with mysterious short-bursts of radio waves or fast radio bursts (FRB) coming from a particular source, for over 10 years since they were discovered in 2007. Fast radio bursts are sudden high-energy bursts



The Hubble Space Telescope is used for SETI research

of astrophysical phenomenon whose origins are unknown. These bursts last for only a few milliseconds and their pulse dispersion measures are much larger for sources present in the Milky Way. The energy generated by them is immensely high, close to the amount of energy generated by the Sun in an entire day. Not all of these FRBs are detected since some of them are so quick that the current devices have a hard time picking them up. Early 2015 is when we actually were quick enough to witness them happening in real time.

There have been a total of 17 FRBs among which one of the signals has been repeating in a sporadic pattern. The first 10 of these signals were detected in March, 2016 but these bursts had actually occurred in May and June 2015. Among the 10, six of them were recorded by the Arecibo radio telescope in Puerto Rico and four of the remaining signals were detected over the course of the next month. Due to a discovery found in older data, the team looked further back and found that a signal was first detected in 2012, making the total count of 11. These FRBs were the first ever ones to be recorded coming outside